INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION
CHAMBER, CALIBRATION

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines requirements for one type of calibration chamber to be used for calibrating various types of atmospheric pressure indicators at the Secondary Reference Level.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6636

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SPECIFICATIONS

MILITARY

MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisional Items (Repair Parts): Packaging of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-970	-	Standards and Specifications, Order of Preference for the Selection of

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Material. The chamber shall be constructed primarily of a corrosion resistant alloy selected in accordance with the order of preference specified in MIL-STD-970. Stainless steel or aluminum may be used at the option of the contractor (see 4.6.1).

3.3 Design and construction. The chamber shall be the shape of a rectangular parallelepiped, with a window in one side and one end removable to provide access. The chamber shall be designed so that the window can be on a vertical side or on top to accommodate the calibration of vertically-and horizontally-read instruments. The design shall orient the chamber with the placement of pads on one side adjacent to the chamber window and on the side opposite the window (see figure 1). Pad material shall be of a suitable material to prevent marking the top of a wooden work bench. The chamber wall

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NOTES:

1. All dimensions in inches with design tolerance of ±0.25 inch.
2. Metric equivalent shall be acceptable.
3. Shape, size, material, and location of PADS; and details of window and door design, is the responsibility of the designer, providing they meet the requirements for this specification.

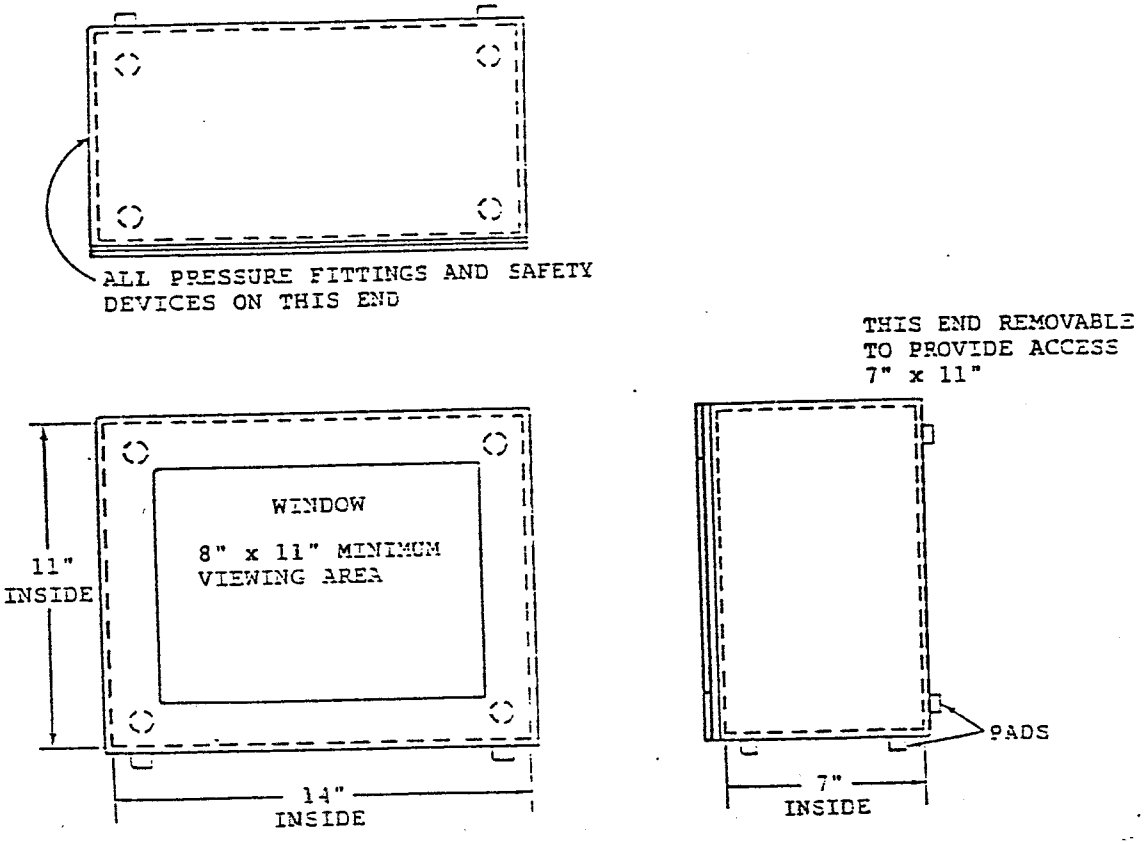


FIGURE 1. Aneroid barometer calibration chamber.

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thickness shall be adequate to meet the stated pressure requirements (see 4.6.2).

3.3.1 Dimensions. The inside dimensions (ID) of the chamber shall be as specified on figure 1 (see 4.6.1).

3.3.2 Weight. The total weight of the chamber shall not exceed 120 pounds (see 4.6.1).

3.3.3 Window. The window shall be centered with respect to the inside dimensions and shall provide a viewing area in either of the larger sides (see figure 1 and 4.6.1). The window shall be sealed with a permanent-type seal. The window shall be safety glass of sufficient thickness to provide an adequate safety factor to withstand the stated positive and negative pressure differentials (see 4.6.2).

3.3.4 Door. The door shall be either of the smaller sides which shall be completely removable to provide access to the chamber interior (see figure 1). The door shall have a reusable seal that can withstand not less than 50 door opening and closing sequences without exceeding the specified leak rate (see 3.4.2). The seal shall be designed to be easily replaceable, requiring no special tools. Design of the door fasteners shall be at the option of the contractor (see 4.6.1).

3.3.5 Pressure fittings. All pressure fittings shall be located on the chamber side opposite the door (see figure 1). The chamber shall have two male tubing connectors (serrated style), each separated from the chamber by a rapid-opening valve such as a toggle valve or ball valve. One of the male tubing connectors shall be suitable for connecting rubber or polyvinyl chloride acetate tubing with 0.25 inch ID and the other shall be suitable for rubber or polyvinyl chloride acetate tubing with a 0.375 inch ID (see 4.6.1).

3.3.6 Pressure relief devices. The chamber shall have a rupture disc, or similar device, to provide adequate protection in the event that a large high-pressure gas source [100 standard cubic feet per minute (scfm) at 100 pounds force/inches squared (lbf/in^2)] is inadvertently introduced. This device should actuate between 8 and 13 pounds per square inch gauge (psig) and not rupture or leak when the chamber is evacuated. The chamber shall have a two-way relief valve (or two relief valves) that opens when the chamber pressure reaches $\pm$ 5 lbf/in^2 of atmospheric pressure. The relief valve(s) must actuate within $\pm$0.5 lbf/in^2 of the specified pressure. All pressure relief devices shall be installed on the chamber side opposite the door (see figure 1, 4.6.2, 4.6.3, and 4.6.4).

3.4 Performance.

3.4.1 Pressure range. Normal operating pressure differential of the chamber shall be $\pm$ 5 lbf/in^2 of atmospheric pressure. The chamber shall be capable of withstanding pressure differentials of 14 lbf/in^2 below atmospheric

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pressure and 20 lbf/in^2 above atmospheric pressure, excluding safety devices, with no damage or distortion (see 4.6.2).

3.4.2 Leak rate. The leak rate shall be not greater than 0.006 cubic centimeters per second (ccs) atmospheric at pressure differentials of ± 4 lbf/in^2 of atmospheric pressure (see 4.6.3).

3.4.3 Pressure relief device calibration. The pressure relief device shall be tested as specified in 4.6.4 for conformance with 3.3.6.

3.5 Identification and marking. An identification plate in accordance with MIL-STD-130 shall be securely affixed to the door of the chamber stating, but not limited to, the following:

Aneroid Barometer Calibration Chamber

Maximum Pressure Differential ± 5 lbf/in^2 of Atmospheric Pressure

Serial No. _____

3.6 Workmanship. Chambers shall be manufactured in such a manner as to be uniform in quality and shall be free from burrs, crazing, cracks, voids, chips, blisters, sharp cutting edges and other defects that will adversely affect life, serviceability, or appearance (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions of 4.3.1 and 4.3.2 of this specification.

4.3.1 Cleaning. Prior to inspection the chamber shall be thoroughly cleaned to remove all foreign matter.

4.3.2 Room temperature and pressure. Unless otherwise specified, all inspections and tests shall be conducted with the chamber at room temperature between 21.1° and 32.2° Celsius (C) and at atmospheric pressure.

4.4 First article inspection. Unless otherwise specified in the contract or order, a first article inspection shall consist of the examination and tests as specified in 4.6 and shall be conducted in the order specified in 4.4.1. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2).

4.4.1 First article sample. The first article sample shall consist of one chamber, representative of the production product. The first article sample shall be subjected to inspections and tests specified in table I in the order shown.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Material	3.2	4.6.1
Examination of product	3.3, 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.6 and 3.6	4.6.1
Low and high pressure	3.4.1	4.6.2
Leak rate	3.4.2	4.6.3
Pressure relief device calibration	3.4.3	4.6.4
Identification and marking	3.5	4.6.1
Packaging	5.1, 5.2	4.6.5

4.5 Quality conformance inspection. The quality conformance inspection shall consist of individual inspection and testing of each chamber.

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4.5.1 Inspection and tests. Each chamber shall be subjected to the inspection and tests specified in table II in the order shown.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Material	3.2	4.6.1
Examination of product	3.3, 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.6 and 3.6	4.6.1
Low and high pressure	3.4.1	4.6.2
Leak rate	3.4.2	4.6.3
Identification and marking	3.5	4.6.1
Packaging	5.1, 5.2	4.6.5

4.6 Methods of inspection.

4.6.1 Examination of product. Each chamber shall be visually and dimensionally inspected and weighed to determine conformance with the requirements of this specification (see 3.2, 3.3, 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.6, 3.5, and 3.6).

4.6.2 Low and high pressure.

4.6.2.1 Low pressure. With the chamber door closed and sealed and pressure relief devices blocked, the chamber shall be evacuated to 14 lbf/in^2 below atmospheric pressure (3.4.1). This pressure shall be held for 5 minutes. There shall be no evidence of damage or distortion.

4.6.2.2 High pressure. With the chamber door closed and sealed and the pressure relief devices blocked, the chamber shall be pressurized to 20 lbf/in^2 above atmospheric pressure (3.4.1). This pressure shall be held for 5 minutes. There shall be no evidence of damage or distortion.

4.6.3 Leak rate.

4.6.3.1 Below atmospheric pressure. With the chamber door closed and sealed, the chamber shall be evacuated to 4 lbf/in^2 below atmospheric pressure and held for 5 minutes. The leakage rate shall be not greater than the requirements of 3.4.2.

4.6.3.2 Above atmospheric pressure. With the chamber door closed and sealed, the chamber shall be pressurized to 4 lbf/in^2 above atmospheric pressure and held for 5 minutes. The leakage rate shall be not greater than the requirements of 3.4.2.

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4.6.4 Pressure relief device calibration.

4.6.4.1 Below atmospheric pressure. With the chamber door closed and sealed, the chamber shall be evacuated to 5 lbf/in^2 below atmospheric pressure. The below atmospheric pressure relief device shall actuate at the pressure requirements of 3.3.6. The procedure shall be repeated five times.

4.6.4.2 Above atmospheric-low pressure. With the chamber door closed and sealed, the chamber shall be pressurized to 5 lbf/in^2 above atmospheric pressure. The above atmospheric-low pressure relief device shall actuate at the pressure requirements of 3.3.6. This procedure shall be repeated five times.

4.6.4.3 Above atmospheric-high pressure. With the chamber door closed and sealed, the chamber shall be rapidly pressurized to 20 lbf/in^2. The above atmospheric-high pressure relief device shall actuate at the pressure requirements of 3.3.6. This procedure shall be repeated five times.

4.6.5 Inspection of preservation, packing and marking. Each chamber prepared for delivery shall be examined to verify compliance with the requirements of 5.1 and 5.2.

5. PACKAGING

5.1 Preservation and packing. Preservation and packing shall be level A, B, or C in accordance with MIL-E-17555.

5.2 Marking. In addition to any special or other identification marking required by the contract (see 6.2) marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The aneroid barometer calibration chamber is intended for use in calibrating atmospheric pressure indicators.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Whether first article sample is required (see 3.1 and 4.4)

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d. Applicable level of preservation and packing required (see section 5)

e. Level of identification and marking required (see 3.5, 5.2)

f. Spare parts as required for each chamber.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting office should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Metering facility
Test environment
Test equipment adjustment procedures
Vacuum testing

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
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